

Design and PCB Implementation of an Electronic Nose System

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Abstract—Electronic nose (e-nose) systems have gained significant attention for applications in environmental monitoring, industrial safety, healthcare, and food quality assessment due to their ability to detect and distinguish multiple gases. This paper presents the design and printed circuit board (PCB) implementation of a compact electronic nose system integrating a fourteen-channel gas sensor array with analog signal conditioning, power management, and embedded processing. The hardware architecture employs an ESP32-S3 microcontroller for system control and wireless communication, while an RP2040 microcontroller is incorporated to generate high-frequency pulse-width modulation (PWM) signals for independent sensor heater control. Dedicated analog front-end circuits were designed to condition the sensor outputs before data acquisition. A rechargeable battery management subsystem, multiple regulated power rails, and environmental sensing were integrated to enable reliable standalone operation. The PCB was designed using KiCad 9.0.7 with careful consideration of component placement, power distribution, ground plane implementation, thermal management, and a detachable perforated sensor board to facilitate sensor replacement and future upgrades. The developed hardware provides a compact and modular platform suitable for multi-gas sensing applications and serves as a foundation for future gas identification and pattern recognition algorithms.

Index Terms—Electronic Nose, Gas Sensors, PCB Design, ESP32-S3, RP2040, Signal Conditioning, KiCad 9.0.7

I. INTRODUCTION

Electronic nose (e-nose) systems have emerged as an effective technology for detecting and distinguishing volatile compounds by employing an array of gas sensors combined with signal processing techniques. Unlike conventional gas detection systems that are designed to identify a single target gas, electronic noses utilize the collective response of multiple sensors to generate characteristic patterns for different gaseous environments. Owing to their versatility, e-nose systems have been widely investigated for applications including environmental monitoring, industrial safety, food quality assessment, agricultural monitoring, and healthcare diagnostics [1]–[3].

Recent advancements in low-power embedded systems and gas sensing technologies have enabled the development of compact and portable electronic nose platforms. In particular, temperature modulation and fast heater cycling have been shown to improve the selectivity and gas discrimination capability of MEMS metal-oxide gas sensors [1]–[3]. However, the integration of multiple gas sensors with their associated analog front-end circuits, heater control, power management, and embedded processing remains a significant hardware design

challenge. Variations in sensor operating conditions, analog signal integrity, power distribution, and thermal effects must be carefully considered during both schematic development and printed circuit board (PCB) layout to ensure reliable system operation.

This work presents the design and PCB implementation of a compact electronic nose system capable of supporting a fourteen-channel gas sensor array. The proposed hardware integrates analog signal conditioning circuits, dedicated heater control, battery management, environmental sensing, and dual microcontroller architecture using the ESP32-S3 and RP2040. The system architecture was designed to provide modularity, efficient power distribution, and simplified sensor integration while maintaining a compact PCB layout suitable for standalone operation.

The PCB was developed using KiCad 9.0.7 with emphasis on component placement, power and ground distribution, thermal management, and the implementation of a detachable perforated sensor board to facilitate sensor replacement and future hardware expansion. The resulting hardware platform provides a flexible foundation for the development of electronic nose applications involving multi-gas sensing and subsequent pattern recognition.

The remainder of this paper is organized as follows. Section II presents the overall system architecture. Section III describes the hardware design, including the gas sensor array, analog signal conditioning circuits, processing units, and power management subsystem. Section IV discusses the PCB layout design and implementation. Section V presents the implementation results and design verification, while Section VI concludes the paper.

II. SYSTEM OVERVIEW

A. Overall System Architecture

Figure 1 illustrates the overall architecture of the proposed electronic nose system. The system integrates a fourteen-channel gas sensor array, analog signal conditioning circuits, dual microcontrollers, power management circuitry, and an environmental sensor on a custom-designed printed circuit board (PCB). The modular architecture was developed to provide reliable sensor interfacing, efficient power distribution, and simplified hardware integration.

The gas sensor array consists of fourteen MEMS-based gas sensors, each interfaced with an analog signal conditioning

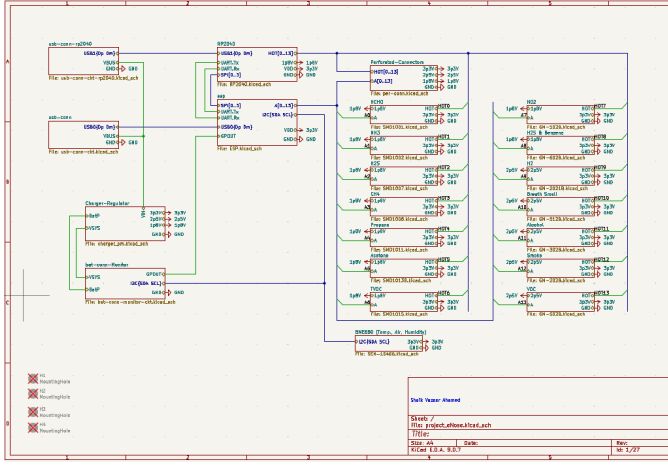


Fig. 1: Overall system architecture.

circuit. The conditioned analog signals from the fourteen MEMS gas sensors are digitized using the ADS131M08 multi-channel analog-to-digital converter (ADC). The digitized sensor data are transferred to the ESP32-S3 through the SPI interface for further processing and wireless communication. An RP2040 microcontroller is incorporated as a dedicated co-processor to generate high-frequency pulse-width modulation (PWM) signals required for independent heater control of the gas sensors. Communication between the two controllers is achieved through standard digital communication interfaces.

The power management subsystem includes a rechargeable battery, battery charging circuit, fuel gauge, and multiple regulated voltage rails to supply the various analog and digital components. In addition, a BME680 environmental sensor is integrated to monitor ambient temperature, humidity, and pressure, enabling environmental compensation during gas sensing. A detachable perforated sensor board is employed to simplify sensor replacement and facilitate future hardware modifications without affecting the main processing board.

B. Overall Architecture

III. HARDWARE DESIGN

A. MEMS Gas Sensor Array

The proposed electronic nose system employs a fourteen-channel MEMS-based gas sensor array to detect a broad range of volatile organic compounds (VOCs) and hazardous gases. MEMS gas sensors were selected due to their compact size, low power consumption, fast thermal response, and improved stability compared with conventional gas sensors. Each sensor incorporates an integrated micro-heater that maintains the sensing material at the required operating temperature.

To enable selective gas detection, sensors with different sensing characteristics were incorporated into the array. The diverse sensor responses provide characteristic signal patterns that can be utilized for subsequent gas identification algorithms. Table I summarizes the gas sensors used in the proposed system and their primary target gases.

TABLE I: MEMS Gas Sensors Used in the Proposed Electronic Nose

Sensor	Target Gas	Heater Voltage
SMD1001	HCHO	1.8 V
SMD1002	NH ₃	1.8 V
SMD1007	H ₂ S	1.8 V
SMD1008	Combustible Gases (CH ₄)	1.8 V
SMD1011	Propane	1.8 V
SMD1013B	Acetone	1.8 V
SMD1015	TVOC	1.8 V
GM-102B	NO ₂	1.8 V
GM-202B	Smoke	2.5 V
GM-2021B	H ₂	2.5 V
GM-302B	Alcohol	2.5 V
GM-502B	VOC	2.5 V
GM-512B	Breath Odor	2.5 V
GM-602B	H ₂ S and Benzene	1.8 V

In addition to the gas sensor array, a BME680 environmental sensor was integrated to measure ambient temperature, humidity, and pressure. These environmental parameters can be used for compensation during sensor calibration and gas sensing, thereby improving measurement consistency under varying operating conditions.

B. Signal Acquisition Circuit

Each MEMS gas sensor is interfaced with an identical signal acquisition circuit to ensure consistent signal conditioning before analog-to-digital conversion. As shown in Fig. 3 and Fig. 4, the sensor output is first passed through a first-order RC low-pass filter with a cutoff frequency of approximately 7 kHz. The filter suppresses high-frequency noise while preserving the low-frequency variations in the sensor response that are relevant for gas detection.

Following the low-pass filter, the conditioned signal is buffered using a TLV9152 rail-to-rail operational amplifier. The amplifier configuration can be selected through solder jumper pads, allowing the circuit to operate either as a voltage follower (unity-gain buffer) or as a non-inverting amplifier with adjustable gain. This configurable design provides flexibility for interfacing sensors with different output voltage ranges without requiring modifications to the PCB.

The operational amplifier also serves as an impedance buffer between the sensor and the analog-to-digital converter (ADC). Its high input impedance minimizes loading of the sensor, while its low output impedance reduces signal degradation caused by long PCB traces and improves the stability of the analog signal presented to the ADS131M08 analog-to-digital converter. The conditioned analog signals are digitized by the ADS131M08 ADC and transferred to the ESP32-S3 through the SPI interface.

C. Processing and Communication

The proposed electronic nose system employs a dual-microcontroller architecture consisting of an ESP32-S3 and an RP2040 to efficiently distribute processing and control

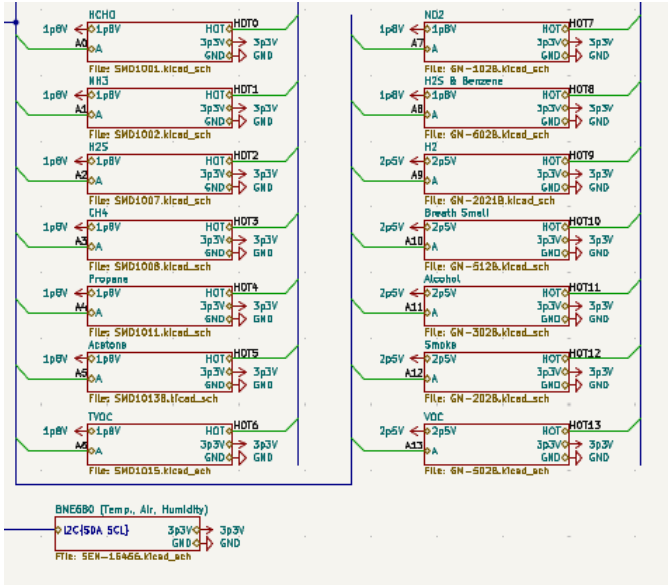


Fig. 2: Arrangement of the fourteen MEMS gas sensors and the BME680 environmental sensor.

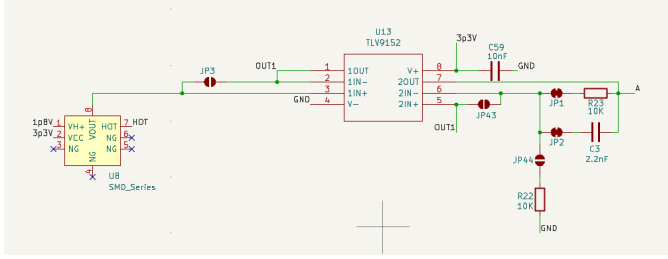


Fig. 3: Representative signal acquisition circuit for SMD-series MEMS gas sensors (7 Sensors).

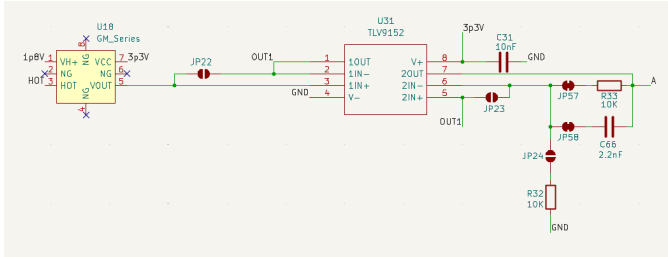


Fig. 4: Representative signal acquisition circuit for GM-series MEMS gas sensors (7 Sensors).

tasks. The ESP32-S3, shown in Fig. 6, serves as the primary processing unit responsible for sensor data acquisition, system control, wireless communication, and peripheral management. The conditioned analog signals from the fourteen MEMS gas sensors are digitized using the ESP32-S3 analog-to-digital converter (ADC) and processed for subsequent analysis.

The RP2040 microcontroller, shown in Fig. 7, is employed as a dedicated co-processor for sensor heater control. It generates high-frequency 3.3 V pulse-width modulation (PWM) control signals, which are applied to the TBD62083AFNG

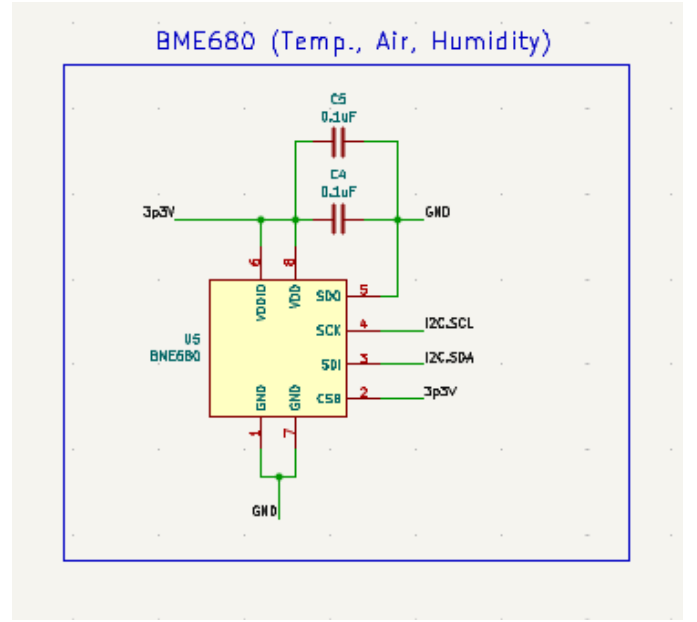


Fig. 5: BME680 Sensor (Pressure, Humidity, and Temperature).

DMOS transistor array. The DMOS driver provides the required current-driving capability to switch the regulated heater supply for the MEMS gas sensors while electrically isolating the microcontroller from the heater load. This dedicated architecture enables reliable heater control without increasing the computational load on the ESP32-S3.

Communication between the ESP32-S3 and RP2040 is established through a UART interface, enabling bidirectional exchange of control commands and system status information. In addition, the ESP32-S3 communicates with peripheral devices through I²C and SPI interfaces, providing support for the BME680 environmental sensor and future hardware expansion. The separation of processing and heater control tasks improves system scalability, simplifies firmware development, and enhances overall system reliability.

D. Power Management

The proposed electronic nose system is powered by a rechargeable lithium-ion battery, enabling portable and standalone operation. A battery charging integrated circuit (IC) is incorporated to support safe charging through a USB interface, while a battery fuel gauge continuously monitors the battery status by measuring parameters such as voltage, current, and remaining capacity. The battery information is communicated to the ESP32-S3, allowing the system to monitor power availability and battery health.

To meet the power requirements of the different hardware modules, dedicated buck converters and voltage regulators generate multiple regulated supply rails from the battery voltage. A 3.3 V supply powers the ESP32-S3, RP2040, BME680 environmental sensor, and other digital peripherals, while separate 2.5 V and 1.8 V supply rails are provided for

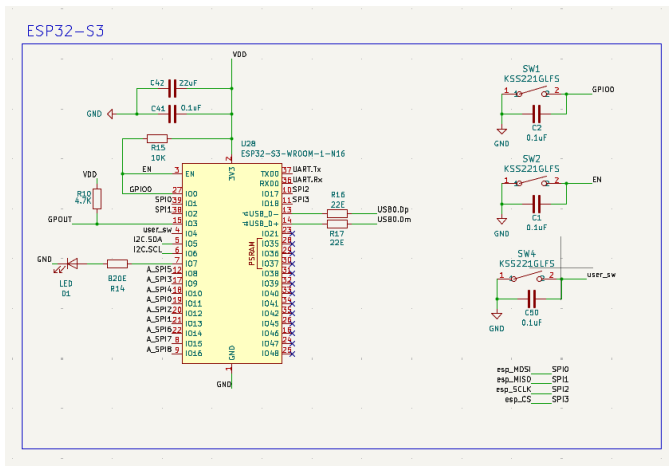


Fig. 6: ESP32-S3 module used as the primary processing unit.

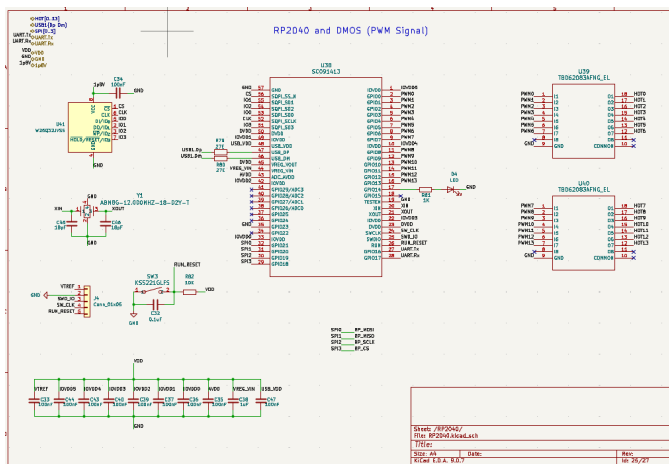


Fig. 7: RP2040 heater-control subsystem with TBD62083AFNG DMOS driver.

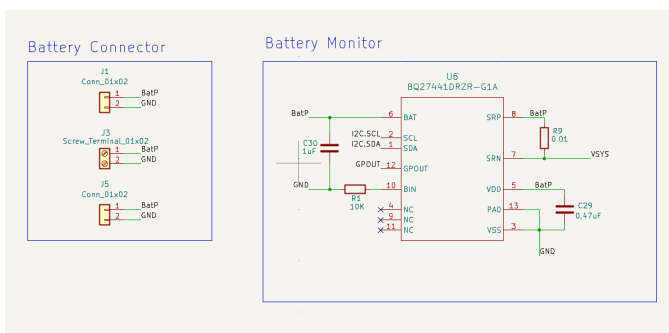


Fig. 8: Battery Connector and fuel gauge circuit.

the MEMS gas sensor heaters based on their operating voltage specifications. This multi-rail power architecture ensures stable operation of all subsystems while improving overall power efficiency. The battery charging circuit supports charging through a USB Type-C interface.

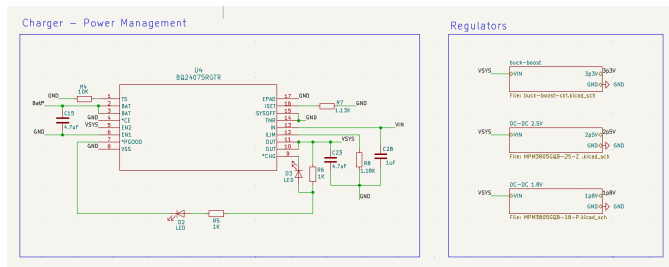


Fig. 9: Battery Charging and Power management circuit showing the regulated power distribution.

IV. PCB LAYOUT DESIGN

The proposed electronic nose system was implemented as a compact single-layer printed circuit board (PCB) using KiCad. The PCB integrates the sensing, signal acquisition, processing, communication, and power management circuits into a single platform while maintaining a layout that minimizes electrical noise and simplifies system assembly. Particular attention was given to component placement, power distribution, and sensor integration to ensure reliable operation of the analog and digital subsystems.

A. Component Placement

The PCB layout was designed by grouping components according to their functionality to improve signal integrity and simplify routing. The fourteen MEMS gas sensors were positioned together on one side of the PCB to facilitate installation inside the gas chamber and to ensure uniform exposure to the target gases. The signal acquisition circuits were placed close to the corresponding sensors to minimize analog trace lengths and reduce noise pickup.

The ESP32-S3 and RP2040 microcontrollers were positioned near the center of the PCB to provide efficient routing to the analog front-end, power management circuitry, and communication interfaces. Battery management components, buck converters, and other power circuits were located away from the sensitive analog circuitry to reduce electrical interference.

B. Sensor Board Integration

To simplify fabrication and future maintenance, the MEMS gas sensors were mounted on a detachable perforated sensor board connected to the main PCB. This arrangement enables straightforward sensor replacement and provides flexibility for evaluating different sensor combinations without redesigning the primary circuit board. The modular configuration also facilitates integration of the sensor array into an enclosed gas chamber while maintaining reliable electrical connections with the signal acquisition circuitry.

C. PCB Implementation

The proposed electronic nose system was implemented as a compact single-layer printed circuit board (PCB) using KiCad. Figure 10 presents the completed PCB layout, highlighting the placement of the MEMS gas sensor array.

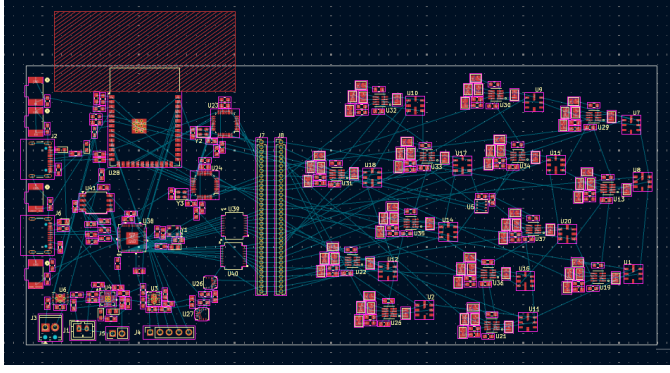


Fig. 10: Top view of the proposed electronic nose PCB layout.

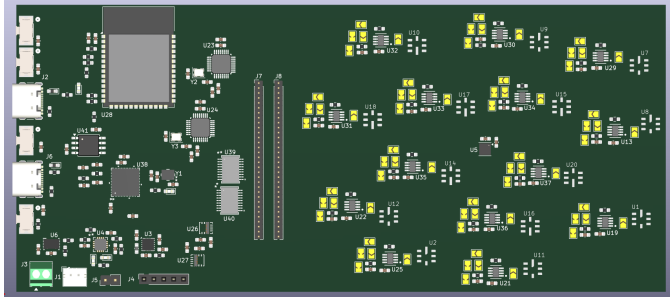


Fig. 11: Three-dimensional view of the proposed electronic nose PCB.

TABLE II: PCB Design Specifications

Parameter	Value
Board Size	73.6 mm x 166.8 mm
Number of Layers	1
Processor	ESP32-S3
PWM Controller	RP2040
Number of Sensors	14
Environmental Sensor	1 (BME680)
PCB Software	KiCad 9.0.7
Supply Voltages	3.3V, 2.5V, 1.8V

signal acquisition circuits, processing units, communication interfaces, and power management circuitry. Figure 11 shows the three-dimensional rendering of the PCB, providing a visual representation of the final hardware implementation. The completed design offers a compact and modular platform suitable for future development and evaluation of electronic nose applications.

V. CONCLUSION

This paper presented the design and PCB implementation of a compact electronic nose system based on a fourteen-channel MEMS gas sensor array. The proposed hardware integrates signal acquisition circuits, a dual-microcontroller architecture consisting of an ESP32-S3 and an RP2040, and a dedicated power management subsystem into a single platform. The RP2040 provides independent PWM-based heater control through a DMOS driver, while the ESP32-S3 performs sensor data acquisition, system control, and communication. The PCB was designed using KiCad with emphasis on modularity, com-

pact component placement, and efficient power distribution. The proposed hardware platform provides a flexible foundation for future implementation of gas classification algorithms and experimental evaluation in electronic nose applications.

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